

# git log/history

## References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

## Git History - Directory Commit Statistics

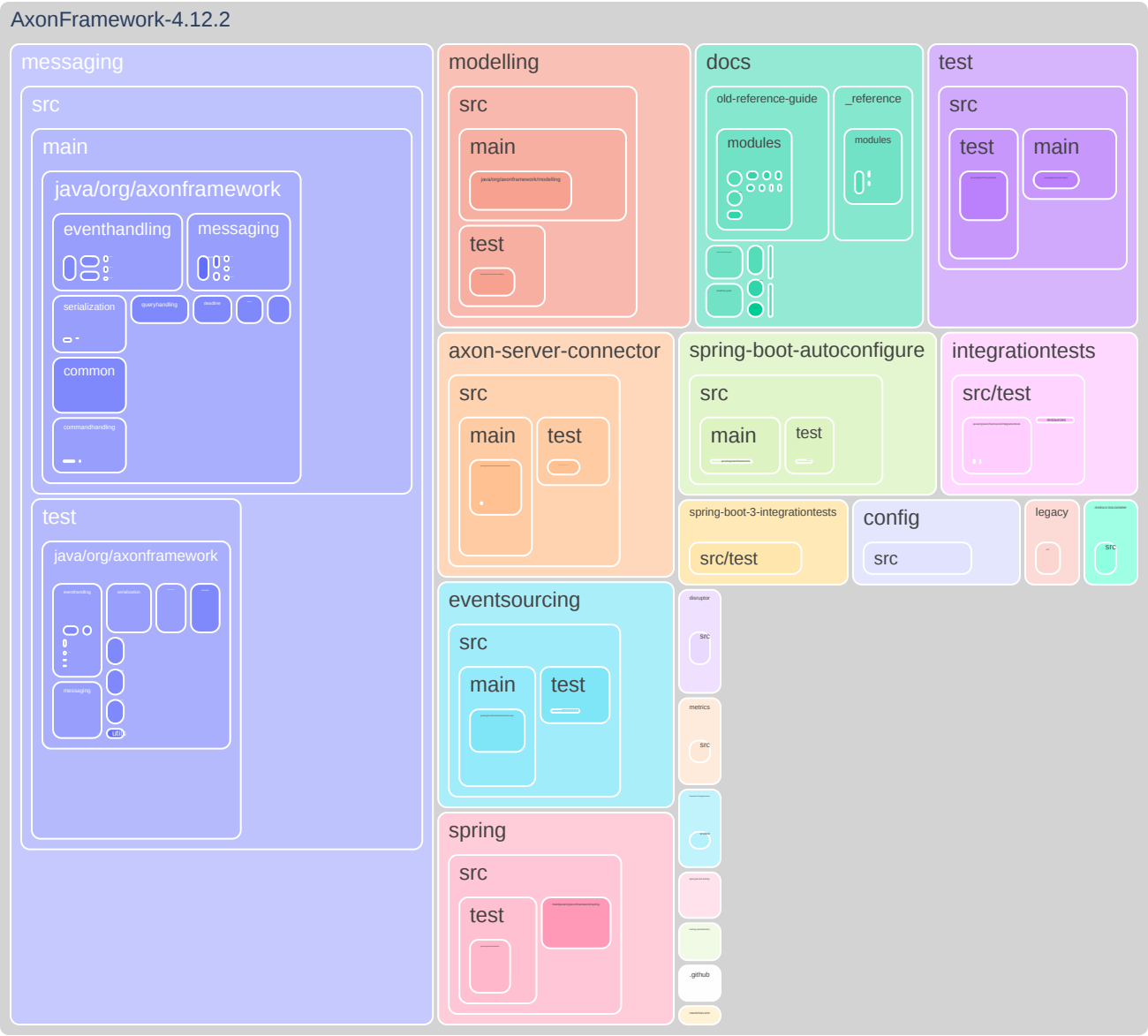
### Data Preview

|       | fileCount   | authorCount | commitCount | daysSinceLastCommit | daysSinceLastCreation | daysSinceLastModification |
|-------|-------------|-------------|-------------|---------------------|-----------------------|---------------------------|
| count | 566.000000  | 566.000000  | 566.000000  | 566.000000          | 566.000000            | 566.000000                |
| mean  | 24.867491   | 19.606007   | 288.137809  | 374.904594          | 1258.489399           | 468.134276                |
| std   | 120.564946  | 21.286048   | 640.533206  | 187.648018          | 982.327748            | 402.673213                |
| min   | 1.000000    | 2.000000    | 3.000000    | 19.000000           | 23.000000             | 18.000000                 |
| 25%   | 1.250000    | 7.000000    | 27.000000   | 249.000000          | 522.000000            | 310.000000                |
| 50%   | 4.000000    | 12.000000   | 81.500000   | 482.000000          | 1041.000000           | 481.000000                |
| 75%   | 10.000000   | 25.000000   | 269.250000  | 482.000000          | 2057.750000           | 481.000000                |
| max   | 2295.000000 | 213.000000  | 9840.000000 | 2632.000000         | 5759.000000           | 3110.000000               |

|    | directoryPath                                                                                                    | directoryParentPath                                                                                       |
|----|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 0  | AxonFramework-4.12.2/assembly                                                                                    | AxonFramework-4.12.                                                                                       |
| 1  | AxonFramework-4.12.2/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/heartbeat/source | AxonFramework-4.12.2/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/heartbeat |
| 2  | AxonFramework-4.12.2/axon-server-connector/src/main/resources/META-INF/services                                  | AxonFramework-4.12.2/axon-server-connector/src/main/resources                                             |
| 3  | AxonFramework-4.12.2/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/event/util       | AxonFramework-4.12.2/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/event     |
| 4  | AxonFramework-4.12.2/config/src/test/resources/META-INF                                                          | AxonFramework-4.12.2/config/src/test/resources                                                            |
| 5  | AxonFramework-4.12.2/coverage-report                                                                             | AxonFramework-4.12.                                                                                       |
| 6  | AxonFramework-4.12.2/docs/_playbook/localLinks                                                                   | AxonFramework-4.12.2/docs/_playbook                                                                       |
| 7  | AxonFramework-4.12.2/docs/_reference/modules/ROOT/attachments                                                    | AxonFramework-4.12.2/docs/_reference/modules/ROOT                                                         |
| 8  | AxonFramework-4.12.2/docs/_reference/modules/ROOT/partials                                                       | AxonFramework-4.12.2/docs/_reference/modules/ROOT                                                         |
| 9  | AxonFramework-4.12.2/docs/_reference/modules/axon-server-connector/pages                                         | AxonFramework-4.12.2/docs/_reference/modules/axon-server-connector                                        |
| 10 | AxonFramework-4.12.2/docs/_reference/modules/axon-server-connector/partials                                      | AxonFramework-4.12.2/docs/_reference/modules/axon-server-connector                                        |
| 11 | AxonFramework-4.12.2/docs/_reference/modules/configuration/pages                                                 | AxonFramework-4.12.2/docs/_reference/modules/configuration                                                |
| 12 | AxonFramework-4.12.2/docs/_reference/modules/configuration/partials                                              | AxonFramework-4.12.2/docs/_reference/modules/configuration                                                |
| 13 | AxonFramework-4.12.2/docs/_reference/modules/disruptor/pages                                                     | AxonFramework-4.12.2/docs/_reference/modules/disruptor                                                    |
| 14 | AxonFramework-4.12.2/docs/_reference/modules/disruptor/partials                                                  | AxonFramework-4.12.2/docs/_reference/modules/disruptor                                                    |
| 15 | AxonFramework-4.12.2/docs/_reference/modules/eventsourcing/pages                                                 | AxonFramework-4.12.2/docs/_reference/modules/eventsourcing                                                |
| 16 | AxonFramework-4.12.2/docs/_reference/modules/eventsourcing/partials                                              | AxonFramework-4.12.2/docs/_reference/modules/eventsourcing                                                |
| 17 | AxonFramework-4.12.2/docs/_reference/modules/legacy/pages                                                        | AxonFramework-4.12.2/docs/_reference/modules/legacy                                                       |
| 18 | AxonFramework-4.12.2/docs/_reference/modules/legacy/partials                                                     | AxonFramework-4.12.2/docs/_reference/modules/legacy                                                       |
| 19 | AxonFramework-4.12.2/docs/_reference/modules/metrics/pages                                                       | AxonFramework-4.12.2/docs/_reference/modules/metrics                                                      |
| 20 | AxonFramework-4.12.2/docs/_reference/modules/metrics/partials                                                    | AxonFramework-4.12.2/docs/_reference/modules/metrics                                                      |
| 21 | AxonFramework-4.12.2/docs/_reference/modules/micrometer/pages                                                    | AxonFramework-4.12.2/docs/_reference/modules/micrometer                                                   |
| 22 | AxonFramework-4.12.2/docs/_reference/modules/micrometer/partials                                                 | AxonFramework-4.12.2/docs/_reference/modules/micrometer                                                   |
| 23 | AxonFramework-4.12.2/docs/_reference/modules/modeling/partials                                                   | AxonFramework-4.12.2/docs/_reference/modules/modeling                                                     |
| 24 | AxonFramework-4.12.2/docs/_reference/modules/spring/pages                                                        | AxonFramework-4.12.2/docs/_reference/modules/spring                                                       |
| 25 | AxonFramework-4.12.2/docs/_reference/modules/spring/partials                                                     | AxonFramework-4.12.2/docs/_reference/modules/spring                                                       |
| 26 | AxonFramework-4.12.2/docs/_reference/modules/springboot-starter/pages                                            | AxonFramework-4.12.2/docs/_reference/modules/springboot-starter                                           |
| 27 | AxonFramework-4.12.2/docs/_reference/modules/springboot-starter/partials                                         | AxonFramework-4.12.2/docs/_reference/modules/springboot-starter                                           |
| 28 | AxonFramework-4.12.2/docs/_reference/modules/test/pages                                                          | AxonFramework-4.12.2/docs/_reference/modules/test                                                         |
| 29 | AxonFramework-4.12.2/docs/_reference/modules/test/partials                                                       | AxonFramework-4.12.2/docs/_reference/modules/test                                                         |

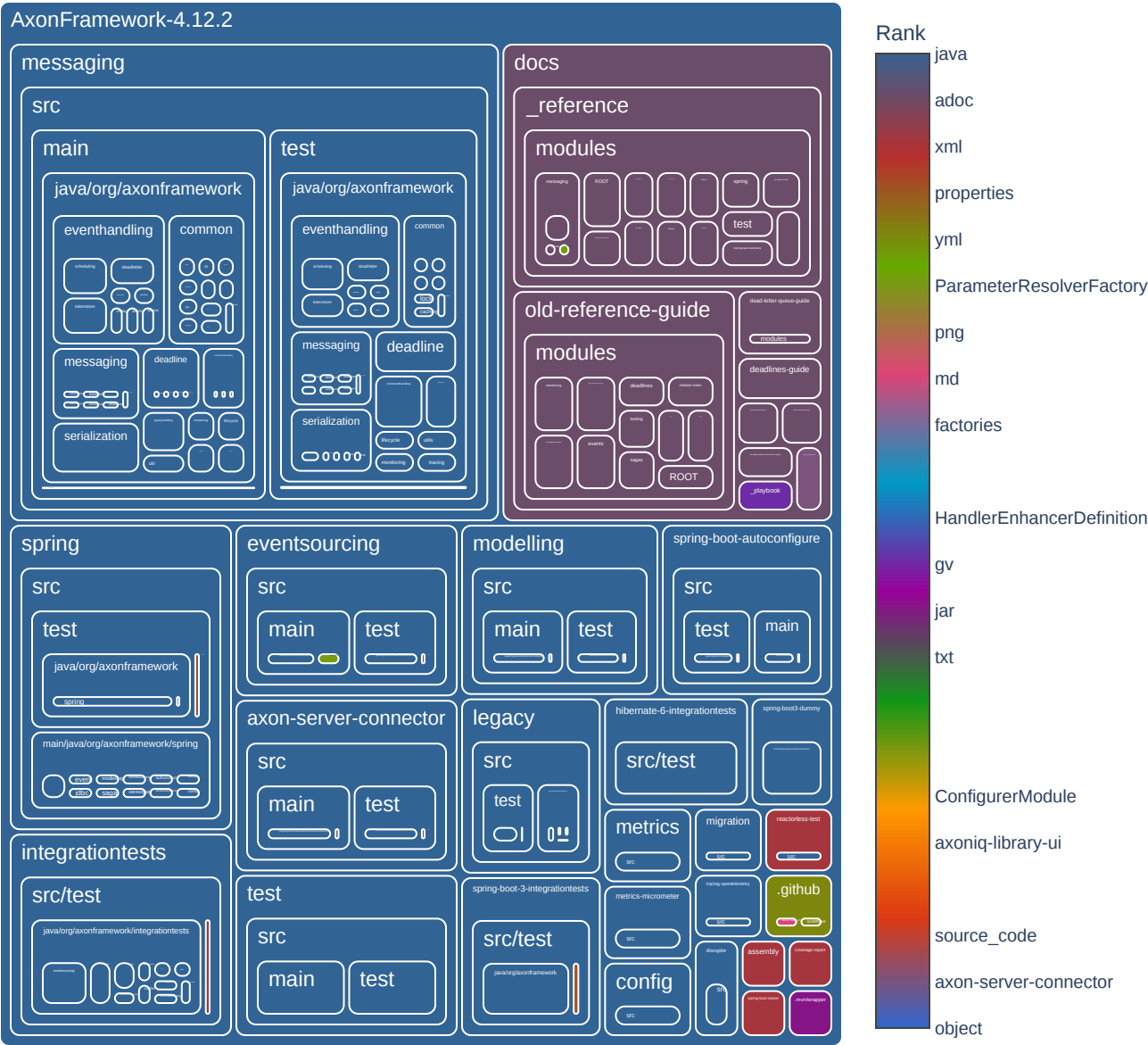
## Number of files per directory

Directories and their file count



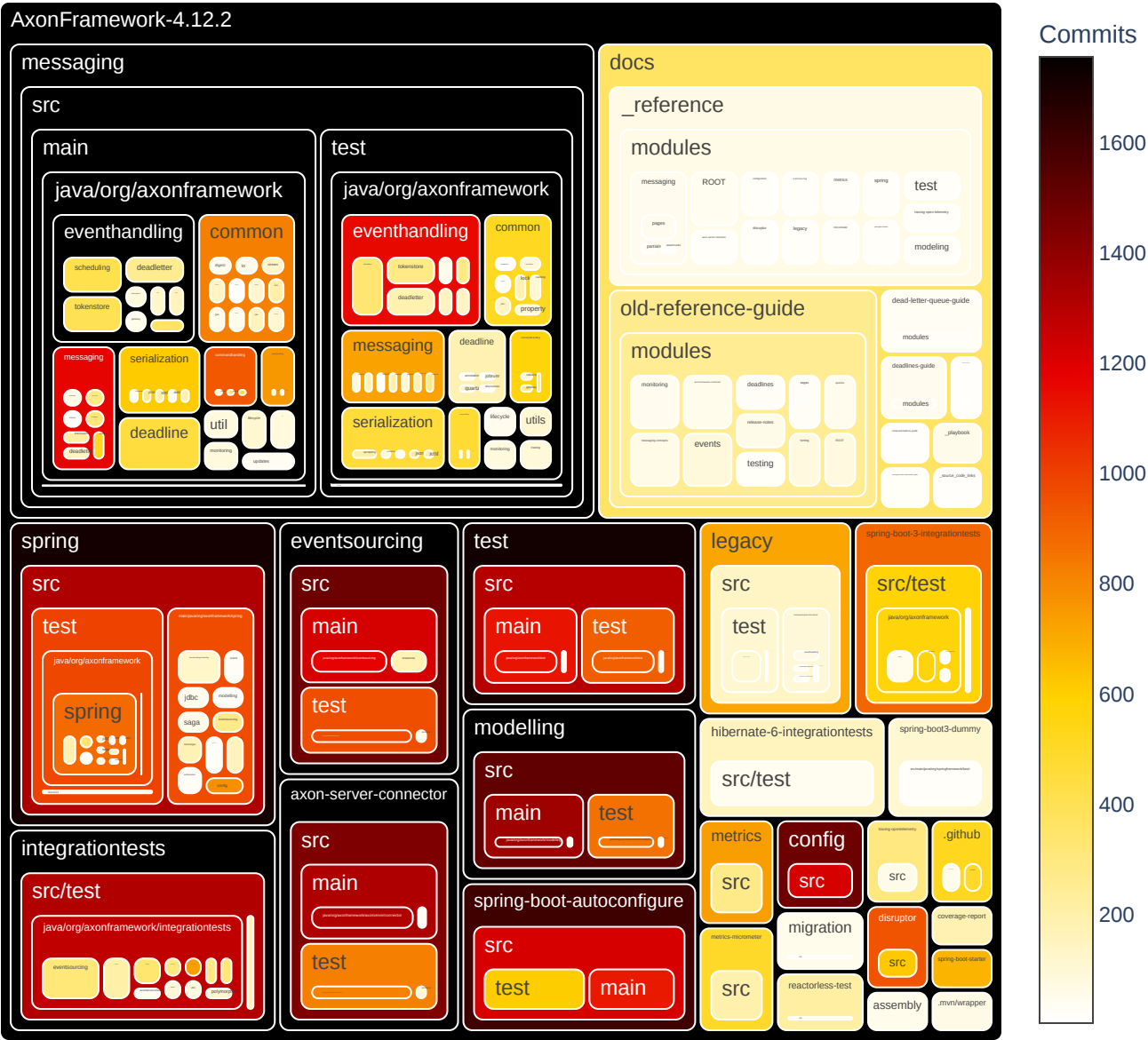
Most frequent file extension per directory

Most frequent file extension per directory



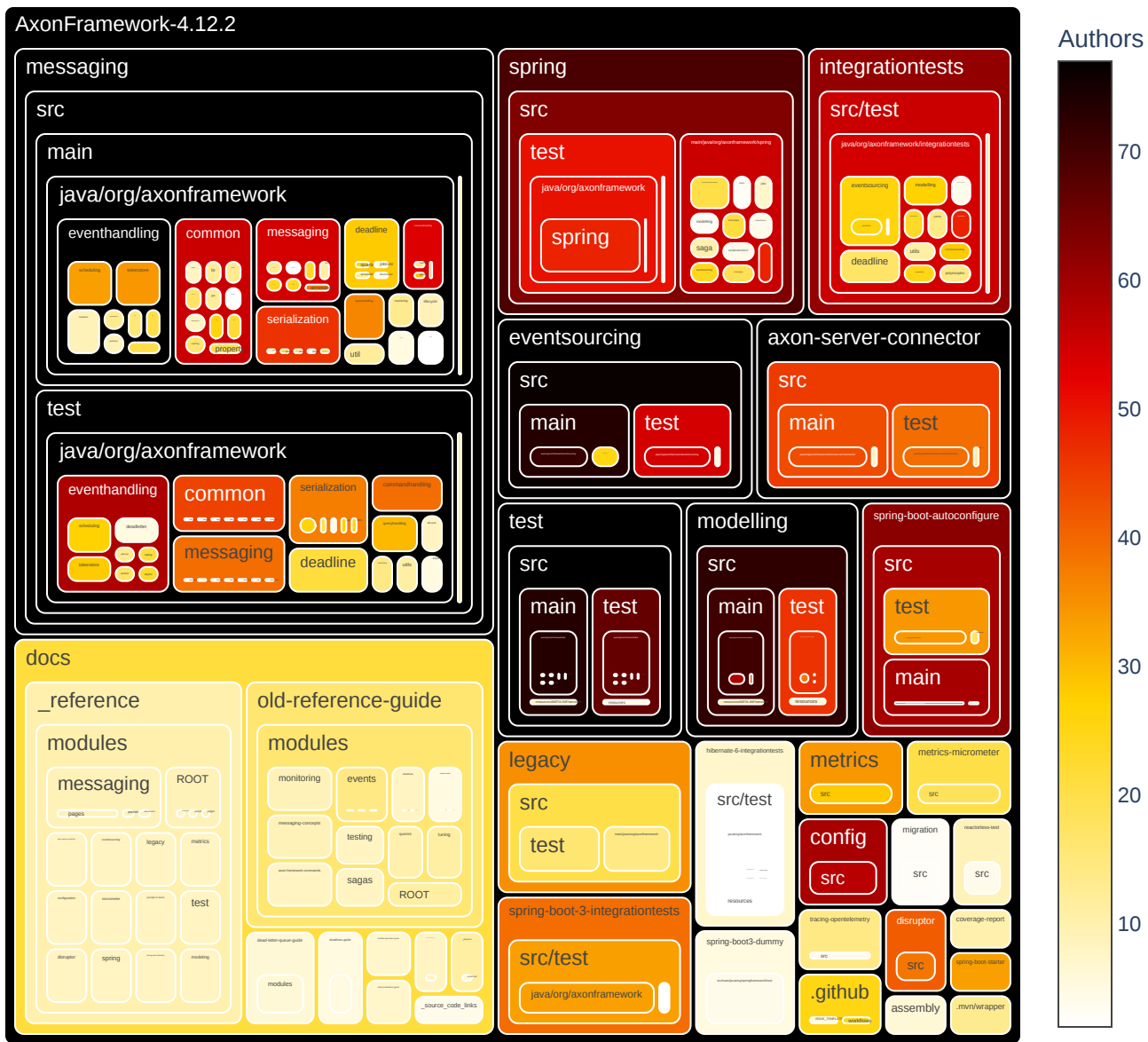
Number of commits per directory

Number of git commits



Number of distinct authors per directory

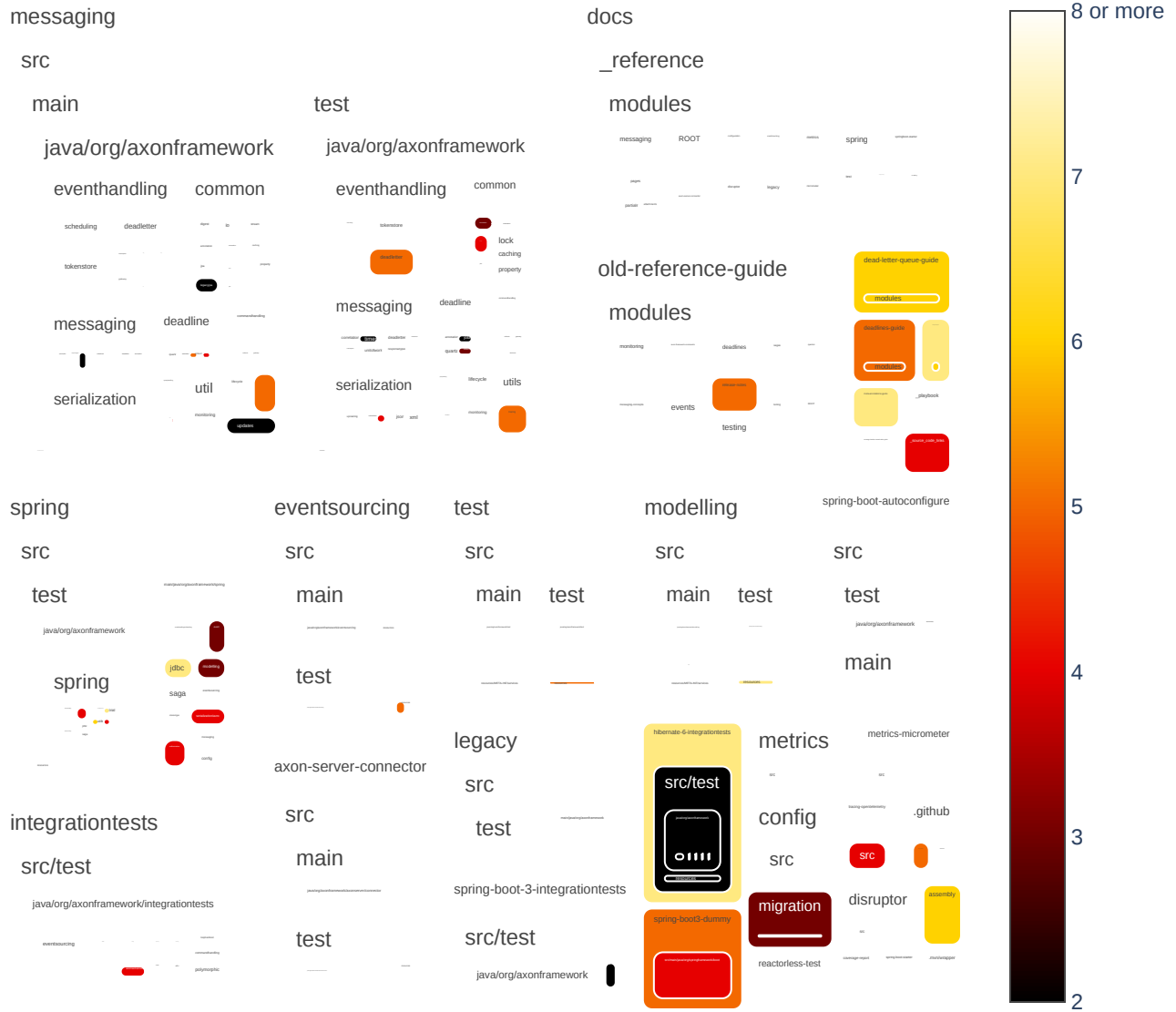
## Number of distinct commit authors



Directories with very few different authors

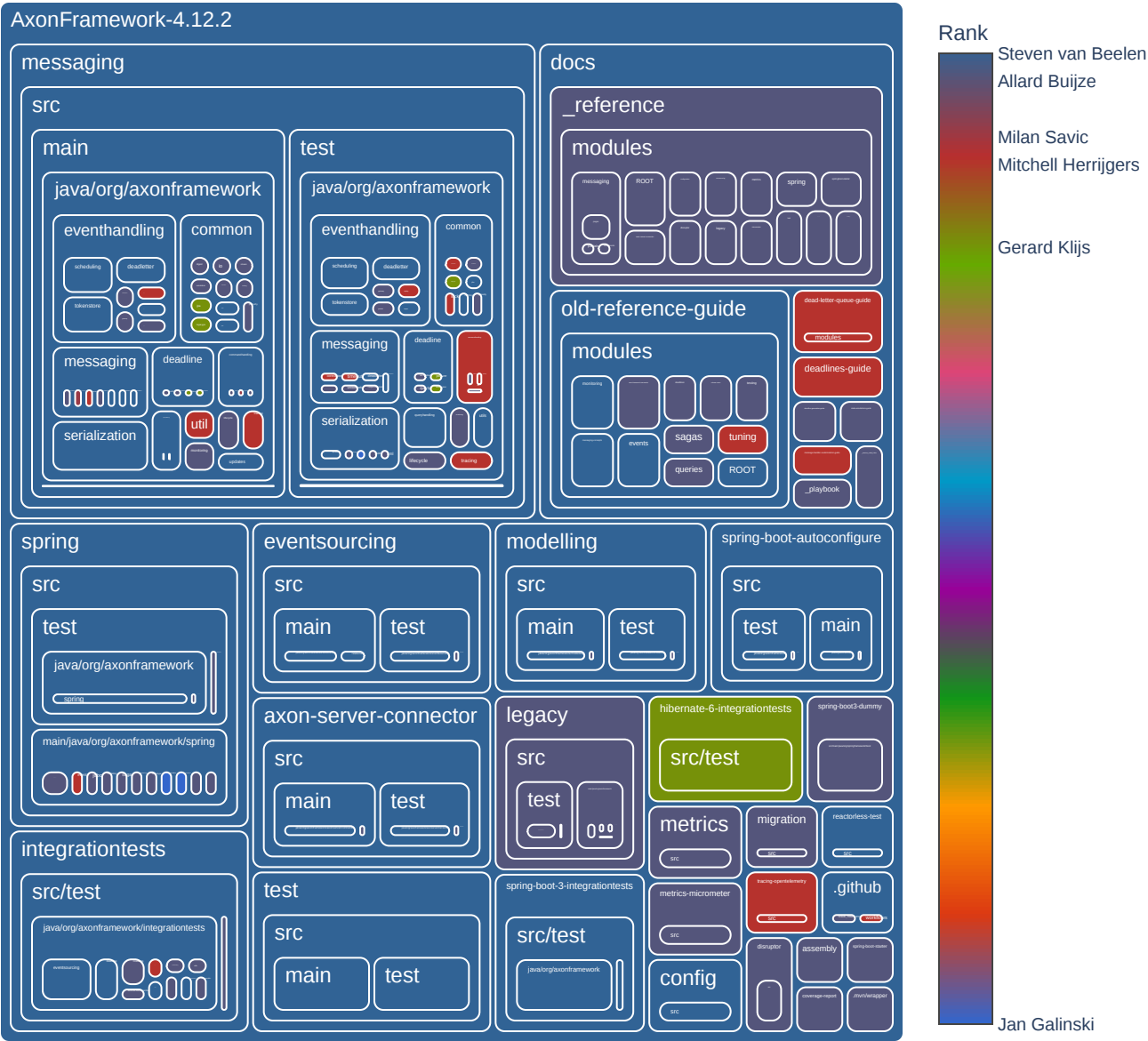
Number of distinct commit authors (red/black = only one or very few authors)

AxonFramework-4.12.2



Main author per directory

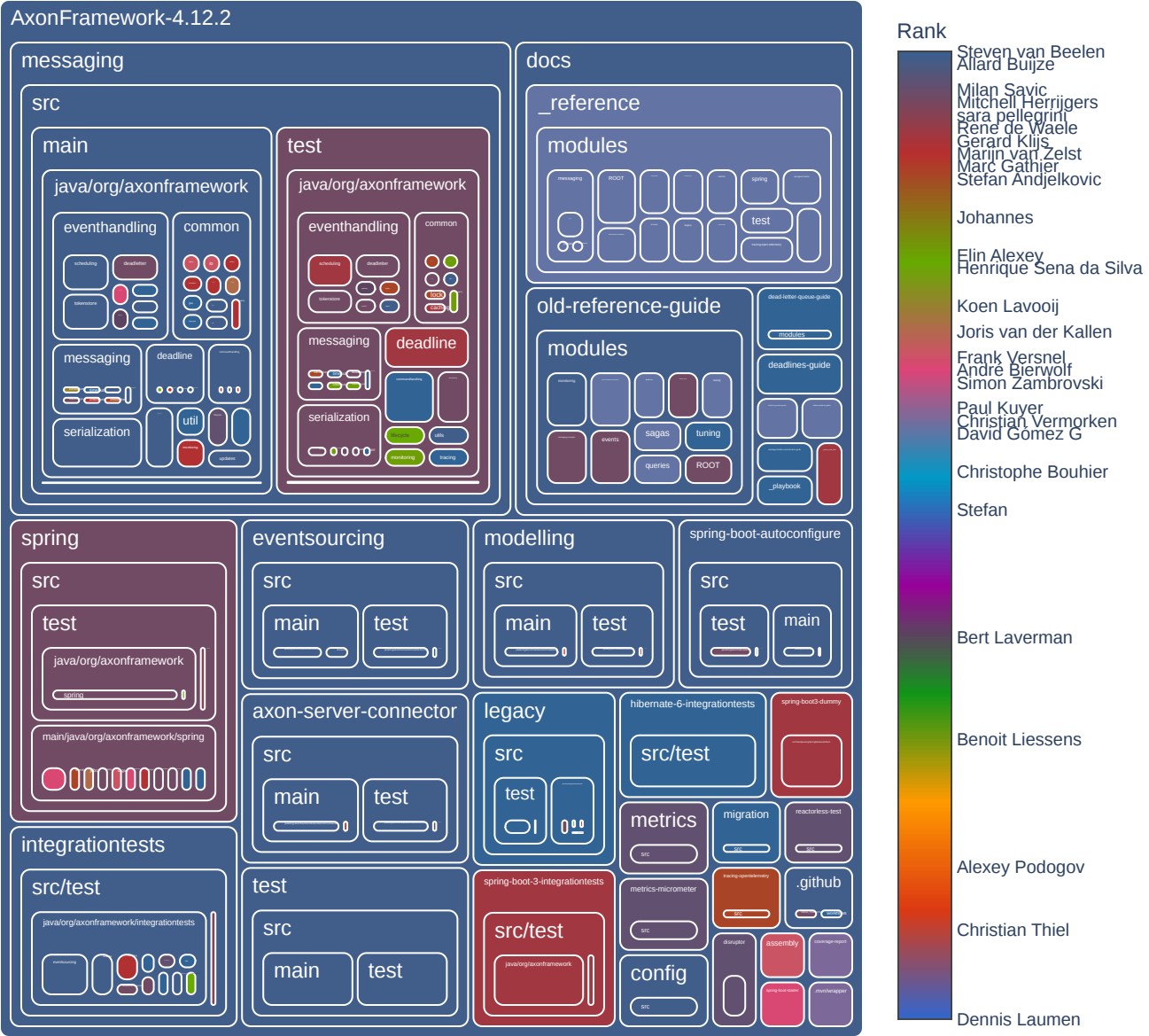
Main authors with highest number of commits



Second author per directory



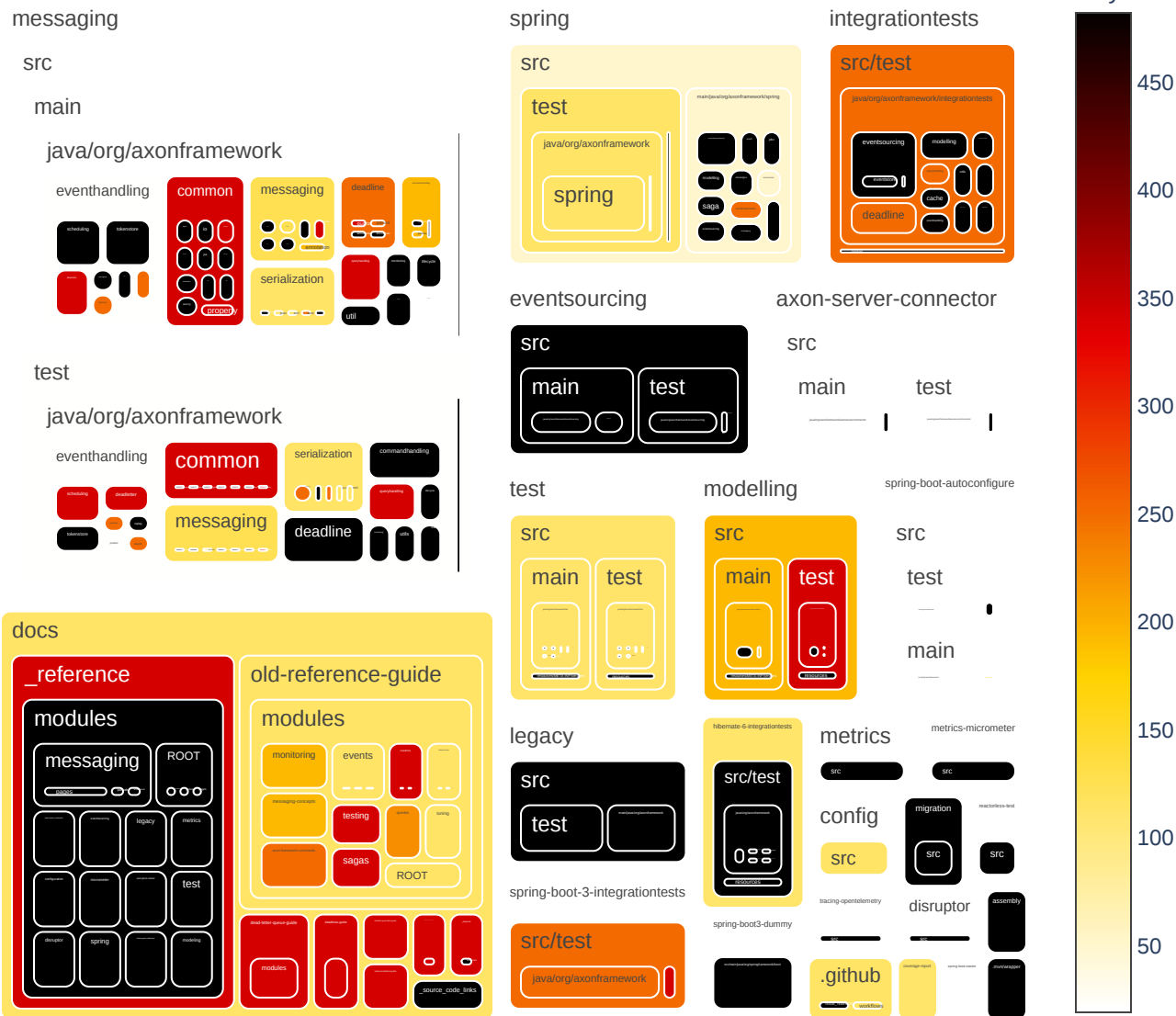
Second author with the second highest number of commits



Days since last commit per directory

Days since last commit

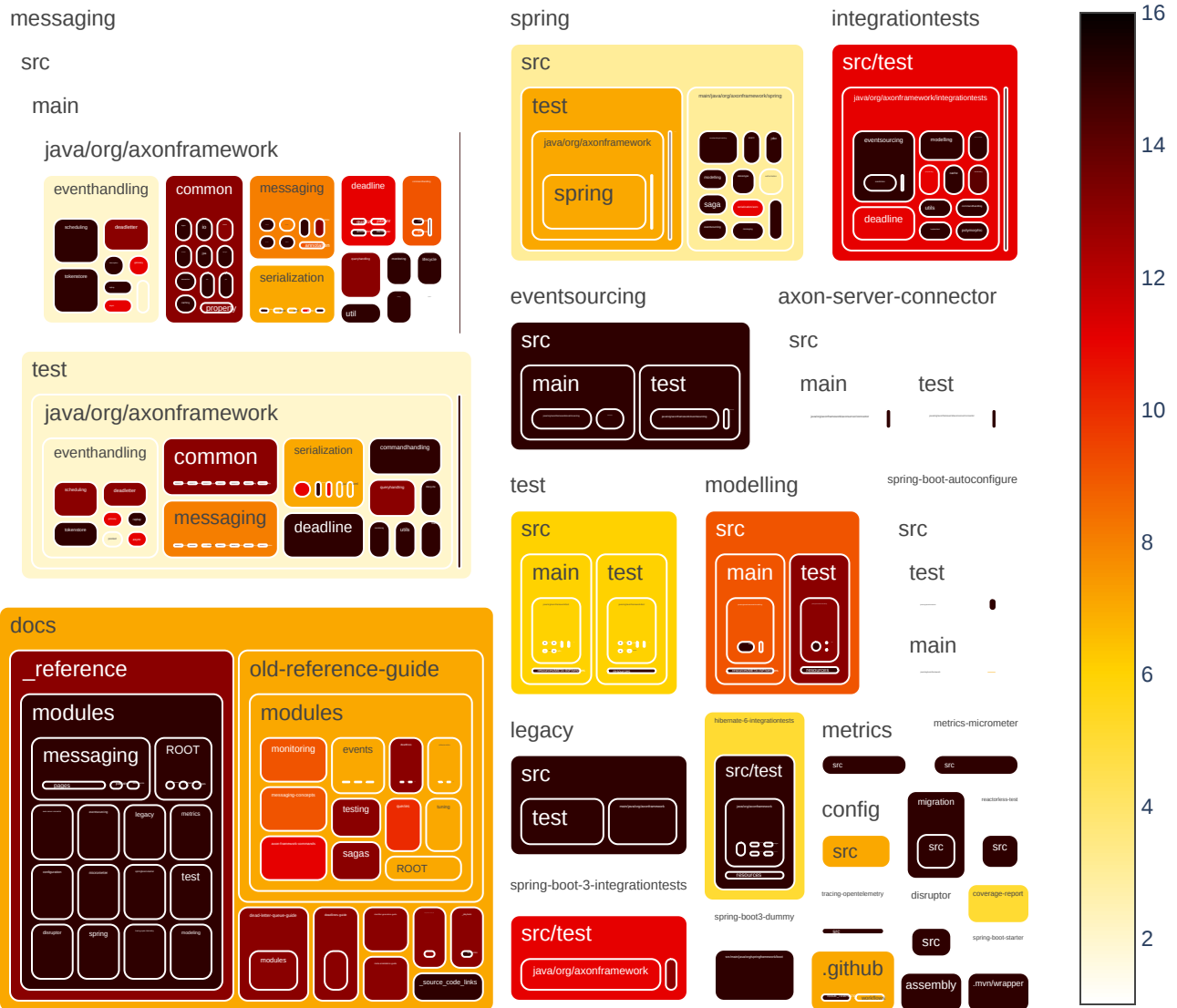
AxonFramework-4.12.2



Days since last commit per directory (ranked)

## Rank of days since last commit

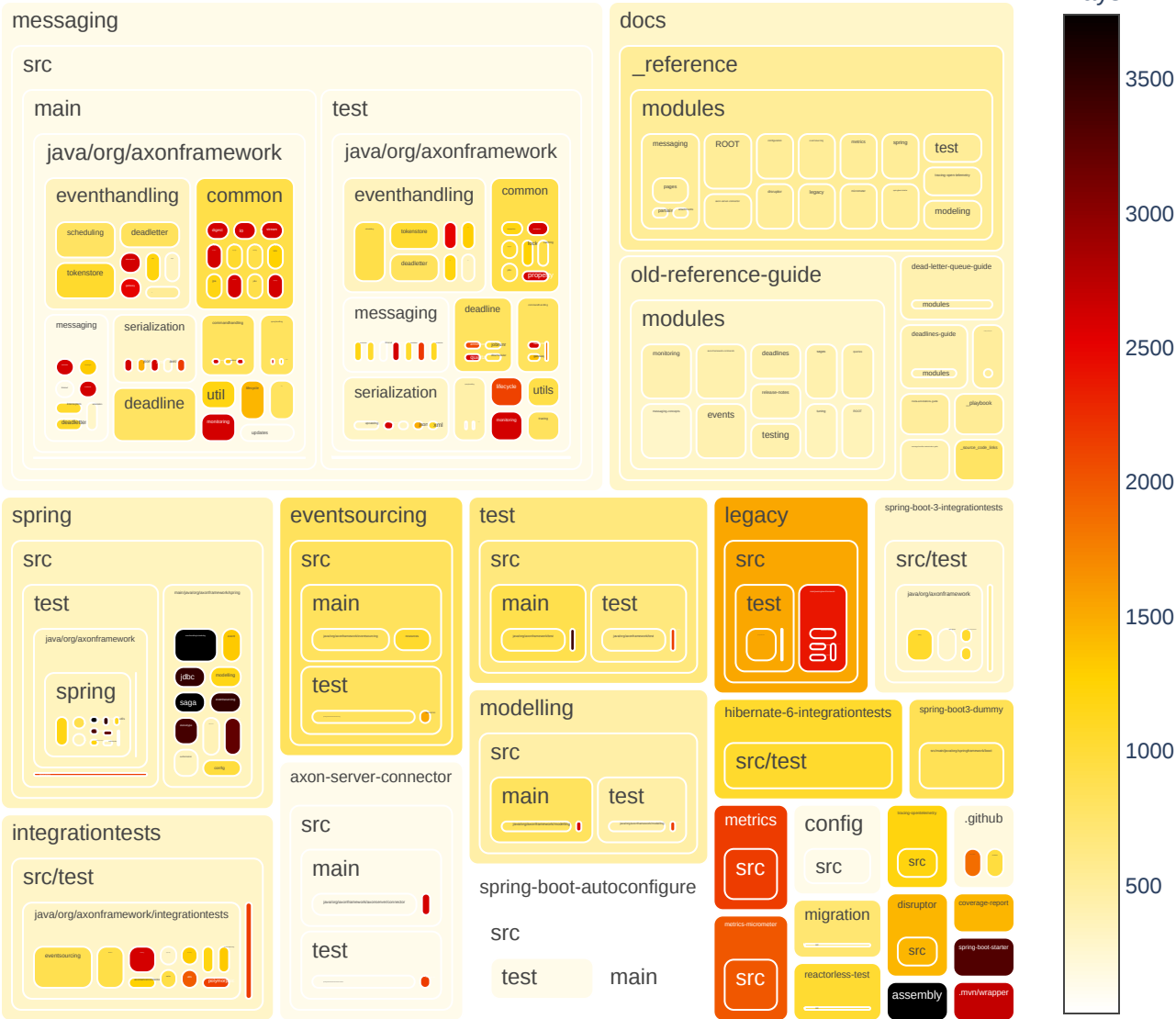
AxonFramework-4.12.2



## Days since last file creation per directory

Days since last file creation

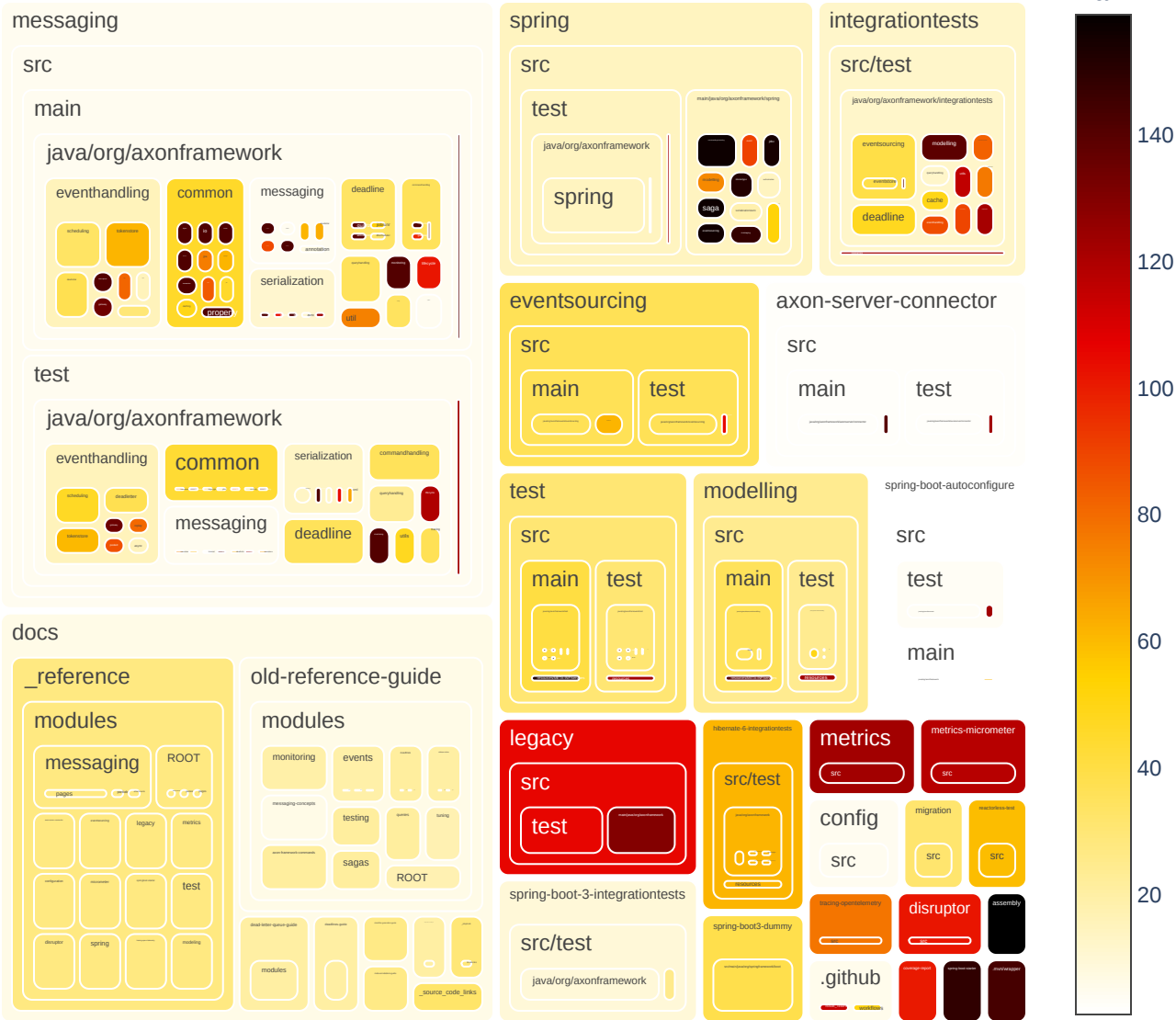
AxonFramework-4.12.2



Days since last file creation per directory (ranked)

Rank of days since last file creation

AxonFramework-4.12.2



Days since last file modification per directory

Days since last file modification

AxonFramework-4.12.2

messaging

src

main

java/org/axonframework



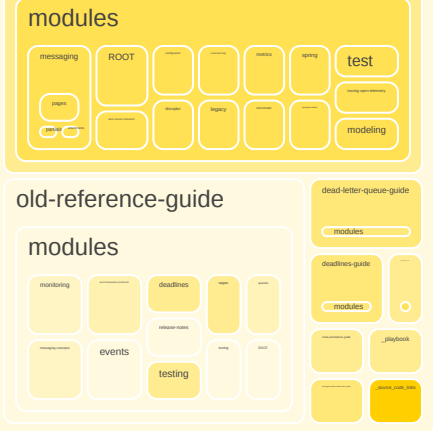
test

java/org/axonframework



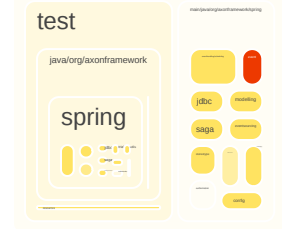
docs

\_reference



spring

src



integrationtests



eventsourcing

src



axon-server-connector

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main

test



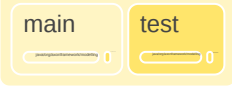
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spring-boot-autoconfigure

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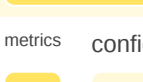
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## Rank of days since last file modification

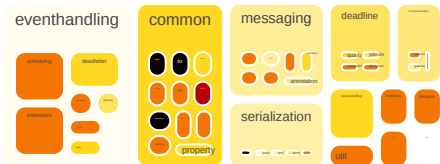
AxonFramework-4.12.2

messaging

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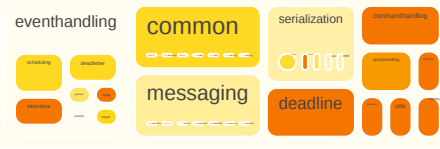
main

java/org/axonframework



test

java/org/axonframework



docs

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modules

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legacy

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spring-boot-3-integrationtests

src/test

java/org/axonframework

integrationtests

src/test

java/org/axonframework/integrationtests

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messaging

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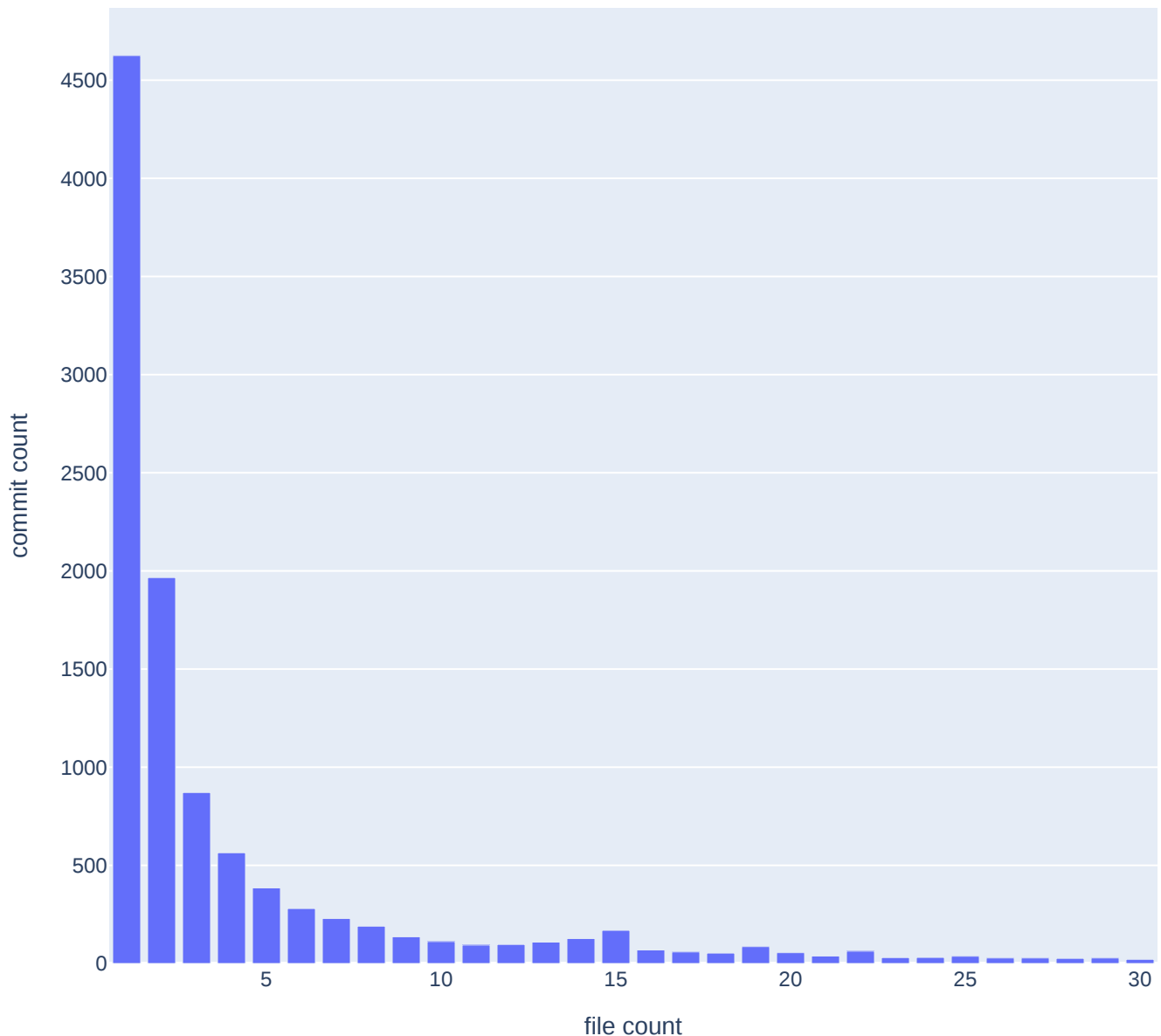
|       | filesPerCommit | commitCount |
|-------|----------------|-------------|
| count | 334.000000     | 334.000000  |
| mean  | 343.449102     | 34.667665   |
| std   | 439.862566     | 281.534044  |
| min   | 1.000000       | 1.000000    |
| 25%   | 84.250000      | 1.000000    |
| 50%   | 185.000000     | 2.000000    |
| 75%   | 426.500000     | 5.000000    |
| max   | 2178.000000    | 4625.000000 |

|    | filesPerCommit | commitCount |
|----|----------------|-------------|
| 0  | 1              | 4625        |
| 1  | 2              | 1966        |
| 2  | 3              | 871         |
| 3  | 4              | 564         |
| 4  | 5              | 385         |
| 5  | 6              | 280         |
| 6  | 7              | 229         |
| 7  | 8              | 190         |
| 8  | 9              | 136         |
| 9  | 10             | 112         |
| 10 | 11             | 94          |
| 11 | 12             | 97          |
| 12 | 13             | 109         |
| 13 | 14             | 127         |
| 14 | 15             | 169         |
| 15 | 16             | 69          |
| 16 | 17             | 59          |
| 17 | 18             | 53          |
| 18 | 19             | 86          |
| 19 | 20             | 55          |
| 20 | 21             | 38          |
| 21 | 22             | 63          |
| 22 | 23             | 30          |
| 23 | 24             | 31          |
| 24 | 25             | 37          |
| 25 | 26             | 28          |
| 26 | 27             | 28          |
| 27 | 28             | 26          |
| 28 | 29             | 28          |
| 29 | 30             | 21          |

Bar chart with the number of files per commit distribution



## Changed files per commit



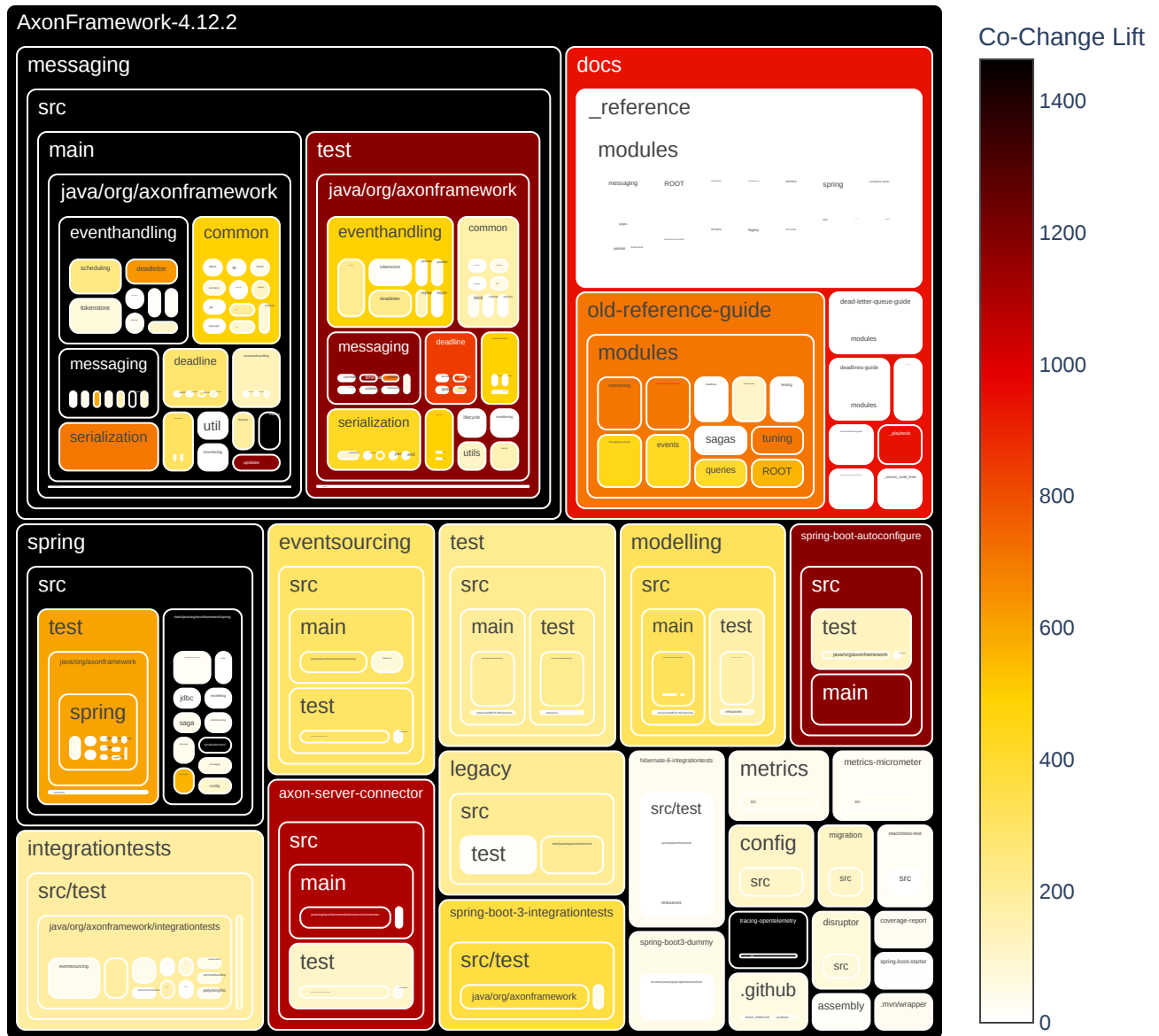
## Pairwise Changed Files

This section analyzes files that were changed together within the same commit and provides several metrics to quantify the strength of the co-change relationship:

- **Commit Count:** The number of commits in which two files were changed together.
- **Commit Lift:** A ratio that indicates whether the co-change pattern is stronger than random chance, given how often each file changes.
- **Jaccard Similarity:** The ratio of commits involving either file that also involved both files.



Co-Changing files in update commits max lift (1=random, >1=more than random, <1=



Co-Changing files in update commits average lift (1=random, >1=more than random,

